

COSC 594/650: Optimization for Machine Learning

Homework 2

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This homework is due on **9/21, 11:59pm**. Format your solution in PDF and submit it on Canvas.

1. (**Quadratic optimization algorithms**) Consider the optimization problem

$$\begin{aligned} & \underset{x}{\text{minimize}} && \frac{1}{2}x^T Qx + b^T x \\ & \text{subject to} && x \in \mathbb{R}^n. \end{aligned}$$

In this homework assignment, we experiment with some of the algorithms introduced in class using the above quadratic optimization problem.

- (a) Generate the matrix Q and the vector b according to the following procedure:
- Set $n = 100$.
 - Generate a random matrix $B \in \mathbb{R}^{n \times n}$ whose entries are drawn from a standard Gaussian distribution $\mathcal{N}(0, 1)$; that is,

$$B = \text{randn}(100, 100).$$

- Generate a diagonal matrix $D \in \mathbb{R}^{n \times n}$ whose diagonal entries are drawn from a standard log-normal distribution; that is,

$$D = \text{diag}(\exp(\text{randn}(n, 1))).$$

- Set $Q = BB^T + D$.
- Generate a random vector b from a zero-mean Gaussian distribution with variance 100; that is, $b = 10 \text{ randn}(n, 1)$.

Solution:

- (b) Using the first-order optimality condition, compute the optimal solution of the above quadratic optimization problem.

Solution:

- (c) Implement the classical gradient descent algorithm with each of the following step-size selection rules:
- i. exact line search;
 - ii. Armijo rule;
 - iii. diminishing step size, $\alpha^r = \alpha/r$; and
 - iv. constant step size, $\alpha^r = 1/L$, where L is the Lipschitz constant of the gradient of the objective function.

Solution:

- (d) Implement the accelerated gradient method discussed in class.

Solution:

- (e) Implement Newton's method.

Solution:

- (f) Start all algorithms from the initial point $x^0 = 0$. Let

$$\varepsilon \triangleq \log \left(\frac{\|x^{1000} - x^*\|}{\|x^0 - x^*\|} \right)$$

be the log-relative accuracy after 1000 iterations.

Solution:

- (g) For each of the above six algorithms, report the average value of ε over five realizations of the problem parameters Q and b .

Solution:

- (h) Report the condition numbers of the five generated realizations.

Solution:

2. **(Effect of condition number)** Repeat Problem 1, but generate the matrix D according to

$$D = \text{diag}(\log(10 + \exp(\text{randn}(n, 1)))) .$$

Notice that this choice of D generally leads to a smaller condition number.

Solution:

3. (Effect of Lipschitz constant estimation)

- (a) Generate one instance of Problem 1.

Solution:

- (b) Plot $\log(e^r)$ versus the iteration number r for both gradient descent with the constant step-size rule and the accelerated gradient descent method, where

$$e^r = \frac{\|x^r - x^*\|}{\|x^0 - x^*\|}$$

is the relative error after r iterations.

Solution:

- (c) Now suppose that the Lipschitz constant is not estimated accurately. Gradually increase the step size from $\alpha = 0.1/L$ to $\alpha = 2.2/L$. For both gradient descent with a constant step size and Nesterov's accelerated method, report the range of step sizes for which the method converges. Which method is more robust to noise in the estimate of the Lipschitz constant?

Solution:

- (d) Repeat the above steps for one instance of Problem 2.

Solution:

4. (Effect of restart in accelerated Nesterov)

- (a) Generate one instance of Problem 2.

Solution:

- (b) Plot $\log(e^r)$ versus the iteration number r for the following algorithms:

- i. Nesterov's accelerated method with no restart;
- ii. Nesterov's accelerated method with restart every $T = \sqrt{\kappa}$ iterations;
- iii. Nesterov's accelerated method with restart every $T = 5\sqrt{\kappa}$ iterations; and
- iv. Nesterov's accelerated method with restart every $T = 20\sqrt{\kappa}$ iterations.

Solution:

- (c) Report your findings in a few sentences.

Solution: